

■ AEGISGUARD
INSTITUTIONAL PORTFOLIO RISK MANAGEMENT PLATFORM

STANDARD OPERATING PROCEDURE
v2.0.0 | RESTRICTED — INTERNAL USE ONLY

Document ID	AG-SOP-2024-001
Classification	CONFIDENTIAL — FIRM EMPLOYEES ONLY
Review Cycle	Quarterly
Owner	Chief Risk Officer
Applies To	Portfolio Management, Risk, Compliance

1. PURPOSE AND SCOPE

AegisGuard is an institutional-grade portfolio risk and optimisation platform designed to enforce systematic, rules-based decision-making across the full investment lifecycle – from asset screening to position exit. This SOP defines the exact workflow that authorised personnel must follow to operate the platform.

Module	Purpose	Who Uses It
Risk Filter Engine	Screens assets for liquidity, vol, correlation	Risk / PM
MVO Optimizer	Maximises Sharpe Ratio within constraints	Portfolio Manager
Regime Detector	Market trend + VIX stance detection	Risk / CIO
Exit Plan Engine	ATR-based trailing stops per asset	Trader / PM
Dashboard	Real-time Bloomberg-style monitoring UI	All Staff

2. WEEKLY OPERATING SCHEDULE

■ MANDATORY: Run the full scan every MONDAY at 09:00 AM (local market time) before any trading activity begins.

Day / Time	Task	Owner	Deadline
MON 09:00	Run full AegisGuard scan	Risk	09:15
MON 09:15	Review Zone classifications	PM	09:30
MON 09:30	Execute any required NO-GO liquidations	Trader	13:30
MON 10:00	Send weekly risk brief to CIO	Risk	10:30
WED 09:00	Mid-week regime check (VIX + SMA)	Risk	09:30
FRI 16:00	Review ATR stops; update exit prices	PM	16:30
DAILY (intraday)	Monitor for CRITICAL EXIT alerts	Trader	Continuous

3. ZONE CLASSIFICATION PROTOCOL

Each asset in the watchlist is automatically classified into one of three zones by the AegisGuard filter engine. The following rules govern actions in each zone.

● GREEN ZONE	Asset passes all three filters: Liquidity Gate, Volatility Cap, and Correlation Matrix. Eligible for MVO optimisation and new long positions. Monitor ATR stops weekly.
--------------	---

● NO-GO ZONE

Asset fails one or more filters. NEW positions are strictly prohibited. Existing positions must be reviewed for liquidation within 4 hours of classification.

● BLACKLIST

Asset triggered the Volatility Cap (20-day vol > 1.5x annual average). Automatic NO-GO override. Full liquidation required within 4 hours. Asset remains blacklisted until vol normalises for 5 consecutive trading days.

4. FILTER DEFINITIONS AND THRESHOLDS

4.1 Liquidity Gate

The Liquidity Gate prevents the firm from placing trades that would move the market against itself. An asset is rejected if:

- Trade Size USD ÷ 20-Day Average Daily Volume (USD) > 1.0%
- Default firm trade size: \$500,000. Adjustable in the sidebar.
- Rationale: A trade exceeding 1% of ADV creates measurable slippage, increasing the effective entry/exit cost.

4.2 Volatility Cap

Protects the portfolio from entering assets undergoing abnormal stress regimes:

- Compute: 20-Day rolling annualised return standard deviation (sigma_short)
- Compute: 252-Day rolling annualised standard deviation (sigma_long)
- REJECT if: $\text{sigma_short} > 1.5 \times \text{sigma_long}$
- Multiplier 1.5 is configurable. Lower values (1.2x) give earlier warnings; higher values (2.0x) tolerate more variance.

4.3 Correlation Matrix

Prevents "fake diversification" – where two assets appear different but move together:

- Compute 252-day pairwise Pearson correlation on log-returns
- When $|\text{rho}(A, B)| > 0.85$, flag the lower-ranked asset for removal
- The surviving asset is the one listed earlier in the watchlist (higher priority)
- Threshold 0.85 is configurable. Institutional standard is 0.75–0.90.

5. PORTFOLIO OPTIMIZATION PROTOCOL

The MVO engine runs Mean-Variance Optimization on all Green Zone assets to find the weight vector w that maximises the Sharpe Ratio:

$$\text{Sharpe} = (w^T \mu - r_f) / \sqrt{w^T \Sigma w}$$

Constraint	Value	Rationale
Sum of weights	= 1.0 (100%)	Fully invested mandate
Individual max weight	<= 20% (default)	Concentration limit (MiFID II / ERISA)

Constraint	Value	Rationale
Short selling	Prohibited (w >= 0)	Long-only institutional mandate
Risk-free rate	4.5% (default)	US T-Bill proxy; adjust to client jurisdiction
Lookback window	252 trading days	1-year rolling parameter estimation

IMPORTANT: The optimizer uses CVXPY (preferred) with SciPy SLSQP as fallback. Always verify the 'Solver Used' field in the Optimization tab post-run. If status shows 'suboptimal', increase the asset universe or relax constraints.

6. MARKET REGIME DETECTION

6.1 Trend Regime (S&P; 500 SMA-200)

Signal	Condition	Required Action
BULL	SPX Close > 200-Day SMA	Normal risk-on positioning
BEAR	SPX Close < 200-Day SMA	Reduce gross equity exposure by >= 25% within 2 trading days
UNKNOWN	Data unavailable	Default to CAUTION posture; escalate to CIO

6.2 VIX Regime (Volatility Index)

VIX Level	Signal	Cash Buffer	Required Action
< 20	NORMAL	5%	Standard allocation; hold cash reserve
20 - 30	ELEVATED	15%	Monitor closely; tighten stop-losses
> 30	DEFENSIVE	30-50%	DEFENSIVE MODE: raise cash, no new longs, tighten ATR stops to 1.5x

■ **DEFENSIVE MODE ACTIVATION:** When VIX exceeds 30, the dashboard will display a red DEFENSIVE MODE banner. The ATR stop multiplier automatically tightens from 2.5x to 1.5x. All new long positions are prohibited until VIX closes below 25 for three consecutive sessions.

7. EXIT PLAN AND TRAILING STOP PROTOCOL

AegisGuard calculates a daily ATR-based exit price for every Green Zone asset. This is the firm's primary downside protection mechanism.

7.1 ATR Calculation

Average True Range (ATR) over 14 periods using Wilder smoothing:

- True Range = max(H-L, |H - Close_prev|, |L - Close_prev|)
- ATR(14) smoothed with Wilder's EMA formula

7.2 Stop Price Calculation

Exit Price = Latest Close - (ATR_Multiplier x ATR14)

Mode	ATR Multiplier	Trigger
Standard	2.5x	Normal regime (VIX < 30)
Defensive	1.5x	VIX > 30 (auto-tightened)

7.3 Mandatory Exit Protocol

RULE: If an asset's current price falls at or below its calculated Exit Price, the AegisGuard dashboard will display a red 'CRITICAL EXIT' badge. The position **MUST** be fully liquidated within 4 hours of market open.

Exit prices are recalculated daily. The trader must check the EXIT PLAN tab each morning before the first trade. Update the firm's order management system with the new stop levels immediately after the Monday scan.

8. ESCALATION AND RESPONSE MATRIX

Event	Severity	Response	Deadline	Escalate To
Asset to NO-GO Zone	HIGH	Halt new positions; begin exit	4 hours	Head of Risk
Asset to BLACKLIST	CRITICAL	Full immediate liquidation	4 hours	CIO + Compliance
VIX > 30 (Defensive Mode)	CRITICAL	Raise cash; tighten stops	Same day	CIO
BEAR Trend Detected	HIGH	Reduce gross exposure >= 25%	2 trading days	CIO + PM
ATR Stop Breached	CRITICAL	Full exit within session	4 hours	Head Trader
Correlation pair flagged	MEDIUM	Flag for Monday rebalancing	Next Monday	PM
All data providers offline	HIGH	Use last cached data; notify Risk	Immediate	Head of Risk
Optimizer returns 'failed'	MEDIUM	Check asset count; review constraints	2 hours	Quant Analyst

9. DATA PIPELINE AND SYSTEM RELIABILITY

9.1 Provider Failover Order

AegisGuard uses a multi-provider data architecture. If the preferred source fails, the system automatically retries the next provider:

Priority	Provider	Cost	Rate Limit
1	Yahoo Finance	Free	~2,000 req/hr
2	AlphaVantage	Freemium	5 req/min (free); 75/min (paid)
3	Polygon.io	Paid	Unlimited (paid tier)

9.2 Local Cache

All fetched data is cached locally in /data/_cache/ as Parquet files. Cache validity period: 4 hours. If a Monday 09:00 scan is run before market open, the system will serve Friday's

cached data – this is expected behaviour. Force a fresh pull using the 'Force Refresh' toggle if available.

9.3 Deployment

Environment	Platform	Access
Production	Streamlit Community Cloud	Firm URL + password auth
Staging	Local (streamlit run app.py)	Developer only
Backup	AWS EC2 or Azure App Service	CTO discretion

10. COMPLIANCE AND AUDIT NOTES

AegisGuard is a decision-support tool. It does not constitute investment advice and does not replace the judgment of qualified investment professionals. All outputs must be reviewed by an authorised person before execution.

- Retain scan reports for a minimum of 7 years (MiFID II / SEC Rule 17a-4).
- All override decisions (ignoring a NO-GO flag) must be documented in writing and countersigned by the CIO within 24 hours.
- Access credentials must be rotated every 90 days.
- The risk-free rate used in Sharpe calculations must reflect the firm's jurisdiction (e.g., ECB rate for EUR-denominated funds).
- This document must be reviewed and re-approved by the CRO every quarter.

11. QUICK REFERENCE – MORNING CHECKLIST

#	Task	Done?
1	Log into AegisGuard dashboard (Monday 09:00)	[]
2	Verify data source status (sidebar – green = live)	[]
3	Enter/confirm watchlist tickers	[]
4	Click RUN FULL SCAN	[]
5	Review OVERVIEW tab – check zone counts	[]
6	Action all NO-GO / BLACKLIST assets (Section 3)	[]
7	Check MARKET REGIME tab – note VIX and SMA-200 signal	[]
8	Check EXIT PLAN tab – update stops in OMS	[]
9	Action any CRITICAL EXIT alerts within 4 hours	[]
10	Send weekly risk brief to CIO by 10:30	[]

DOCUMENT APPROVAL

Role	Name	Signature	Date
Chief Risk Officer			
Portfolio Manager			
Compliance Officer			
Chief Technology Officer			

This document is the property of the firm. Any reproduction or distribution outside the firm without written authorisation from the CRO is strictly prohibited.